Lesson 2.4 Confunding in the 3^k Factorial Design

Introduction

Even when a single replicate of the 3k design is considered, the design requires so many runs that it is unlikely that all 3k runs can be made under uniform conditions. Thus, confounding in blocks is often necessary. The 3k design may be confounded in 3p incomplete blocks, where $p < k$. Thus, these designs may be confounded in three blocks, nine blocks, and so on.

The 3^k Factorial Design in Three Blocks

Suppose that we wish to confound the 3^k design in three incomplete blocks. These three blocks have two degrees of freedom among them; thus, there must be two degrees of freedom confounded with blocks. Recall that in the 3^k factorial series each main effect has two degrees of freedom. Furthermore, every two-factor interaction has four degrees of freedom and can be decomposed into two components of interaction (e.g., AB and AB^2), each with two degrees of freedom; every three-factor interaction has eight degrees of freedom and can be decomposed into four components of interaction (e.g., ABC, ABC^2, AB^2C, and AB^2C^2), each with two degrees of freedom; and so on. Therefore, it is convenient to confound a component of interaction with blocks.

The general procedure is to construct a defining contrast

$$L = \alpha_1 x_1 + \alpha_2 x_2 + \cdots + \alpha_k x_k$$

where α_i represents the exponent on the ith factor in the effect to be confounded and x_i is the level of the ith factor in a particular treatment combination. For the 3^k series, we have $\alpha_i = 0$, 1, or 2 with the first nonzero α_i being unity, and $x_i = 0$ (low level), 1 (intermediate level), or 2 (high level). The treatment combinations in the 3^k design are assigned to blocks based on the value of $L(mod3)$. Because $L(mod3)$ can take on only the values 0, 1, or 2, three blocks are uniquely defined. The treatment combinations satisfying $L = 0(mod3)$ constitute the principal block. This block will always contain the treatment combination $00 \cdots 0$.

For example, suppose we wish to construct a 3^2 factorial design in three blocks. Either component of the AB interaction, AB or AB^2, may be confounded with blocks. Arbitrarily choosing AB^2, we obtain the defining contrast

$$L = x_1 + 2x_2$$

The value of L (mod 3) of each treatment combination may be found as follows:

$$
\begin{aligned}
00 &: L = 1(0) + 2(0) = 0(mod3) = 0 \\
01 &: L = 1(0) + 2(1) = 2(mod3) = 2 \\
02 &: L = 1(0) + 2(2) = 4(mod3) = 1 \\
10 &: L = 1(1) + 2(0) = 1(mod3) = 1 \\
11 &: L = 1(1) + 2(1) = 3(mod3) = 0 \\
12 &: L = 1(1) + 2(2) = 5(mod3) = 2 \\
20 &: L = 1(2) + 2(0) = 2(mod3) = 2 \\
21 &: L = 1(2) + 2(1) = 4(mod3) = 1 \\
22 &: L = 1(2) + 2(2) = 6(mod3) = 0
\end{aligned}
$$

The blocks are shown below.

Block 1	Block 2	Block 3
00	10	01
11	21	12
22	02	20

The elements in the principal block form a group with respect to addition modulus 3. Referring to the figure, we see that $11 + 11 = 22$ and $11 + 22 = 00$. Treatment combinations in the other two blocks may be generated by adding, modulus 3, any element in the new block to the elements of the principal block. Thus, we use 10 for block 2 and obtain

An example

We illustrate the statistical analysis of the 3^2 design confounded in three blocks by using the following data, which come from the single replicate of the 3^2 design.

	Block 1	Block 2	Block 3
	00 = 4	10 = −2	01 = 5
	11 = −4	21 = 1	12 = −5
	22 = 0	02 = 8	20 = 0
Block totals =	0	7	0

Using conventional methods for the analysis of factorials we get the following sums of squares.

```r
library(tidyverse)
library(kableExtra)
Block <- c(rep(1,3), rep(2,3),rep(3,3))
A <- c( 0, 1, 2, 1, 2, 0, 0, 1, 2)
B <- c(0, 1, 2, 0, 1, 2, 1, 2, 0)
Y <- c( 4, -4, 0, -2, 1, 8, 5, -5, 0)
df <- data.frame(cbind(Block, A, B, Y))
df <- df %>%
  mutate(Block = factor(Block),
         A = factor(A),
         B = factor(B))
out1 <- aov(Y ~ Block + A*B, data = df)
kable(anova(out1), digits = 4)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Block	2	10.8889	5.4444	NaN	NaN
A	2	131.5556	65.7778	NaN	NaN
B	2	0.2222	0.1111	NaN	NaN
A:B	2	2.8889	1.4444	NaN	NaN
Residuals	0	0.0000	NaN	NA	NA

However, SS_{Blocks} is exactly equal to the AB^2 component of interaction. To see this, write the observations as follows:

		Factor *B*		
		0	**1**	**2**
	0	4	5	8
Factor *A*	1	−2	−4	−5
	2	0	1	0

Recall from the previous lesson that the $I(AB)$ or AB^2 component of the AB interaction may be found by computing the sum of squares between the left-to-right diagonal totals in the above layout. This yields

$$SS_{AB^2} = \frac{0^2 + 0^2 + 7^2}{3} - \frac{7^2}{9} \approx 10.889$$

which is identical to SS_{Blocks}.

Confounding in a 3^3 design

We now look at a slightly more complicated design—a 3^3 factorial confounded in three blocks of nine runs each. The AB^2C^2 component of the three-factor interaction will be confounded with blocks. The defining contrast is

$$L = x_1 + 2x_2 + 2x_3$$

It is easy to verify that the treatment combinations 000, 012, and 101 belong in the principal block. The remaining runs in the principal block are generated as follows:

$$\begin{aligned}
012 + 012 &= 021 \\
101 + 101 &= 202 \\
101 + 012 &= 110 \\
101 + 021 &= 122 \\
012 + 202 &= 211 \\
021 + 202 &= 220
\end{aligned}$$

To find the runs in another block, note that the treatment combination 200 is not in the principal block. Thus, the elements of block 2 are

$$\begin{aligned}
200 + 000 &= 200 \\
200 + 102 &= 212 \\
200 + 101 &= 001 \\
200 + 021 &= 221 \\
200 + 202 &= 102 \\
200 + 110 &= 010 \\
200 + 122 &= 022 \\
200 + 211 &= 111 \\
200 + 220 &= 120
\end{aligned}$$

The blocks are shown in the figure below. Verify the runs in Block 3.

Block 1	Block 2	Block 3
000	200	100
012	212	112
101	001	201
202	102	002
021	221	121
110	010	210
122	022	222
211	111	011
220	120	020